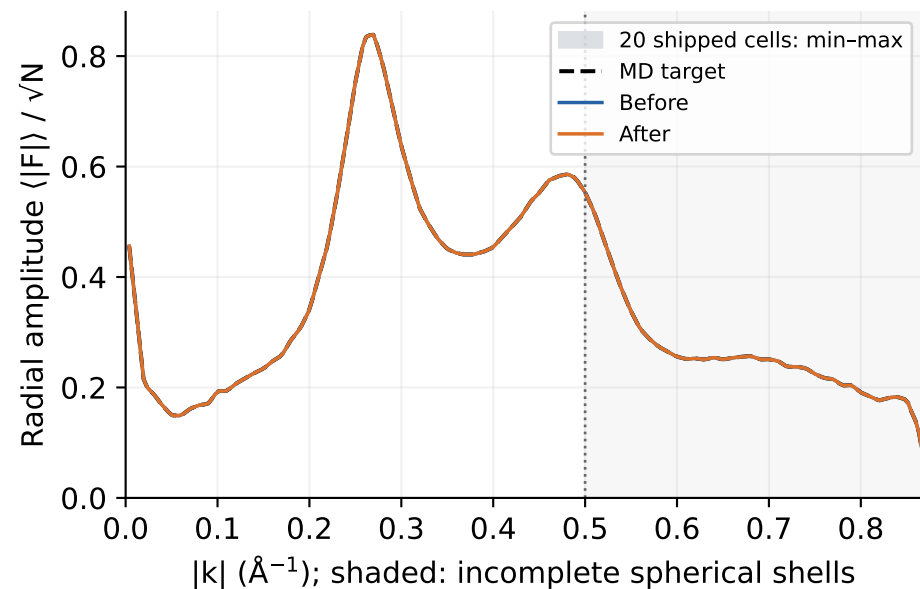
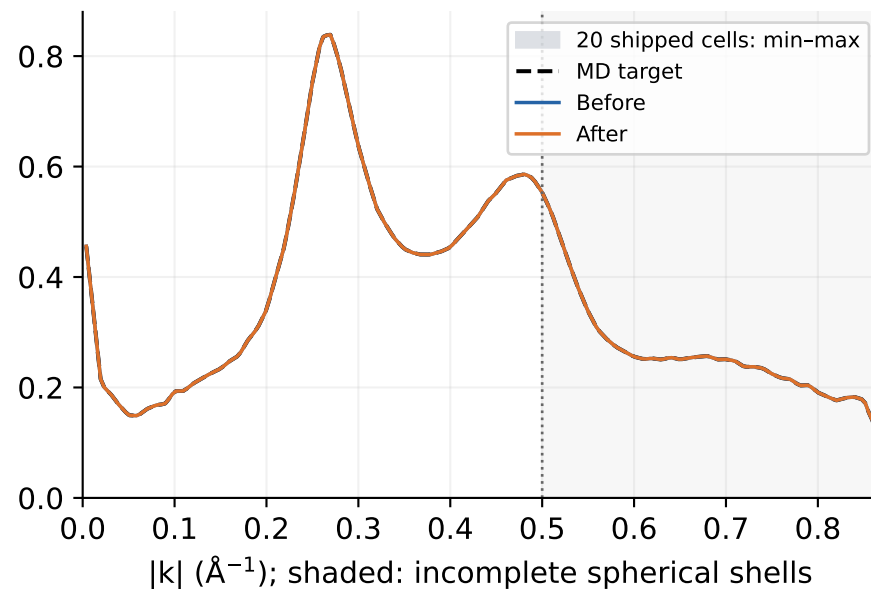


Before / after Fourier spectra of the saved production ice
256³ at 1 Å/voxel · 527,178 molecules · DC omitted · residuals magnified below

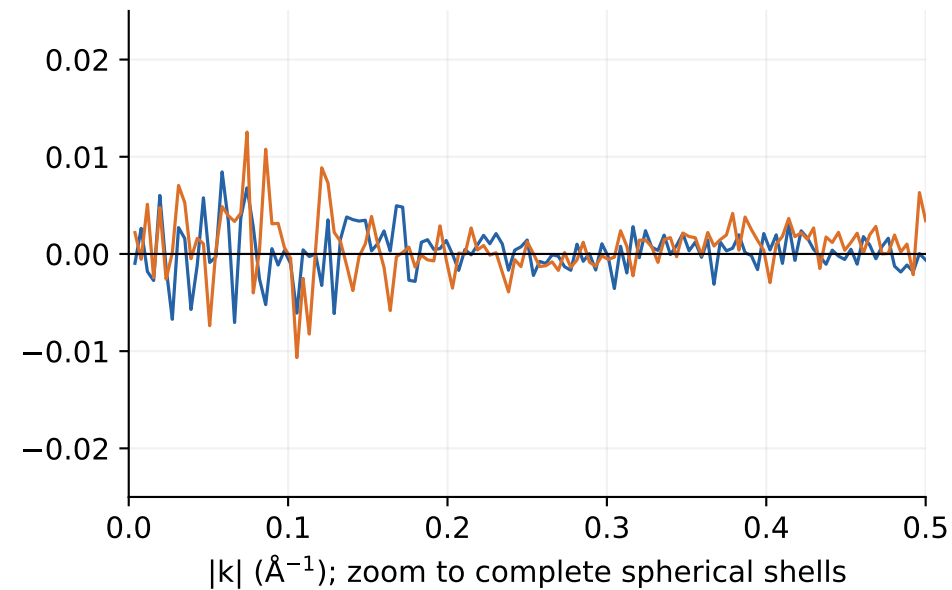
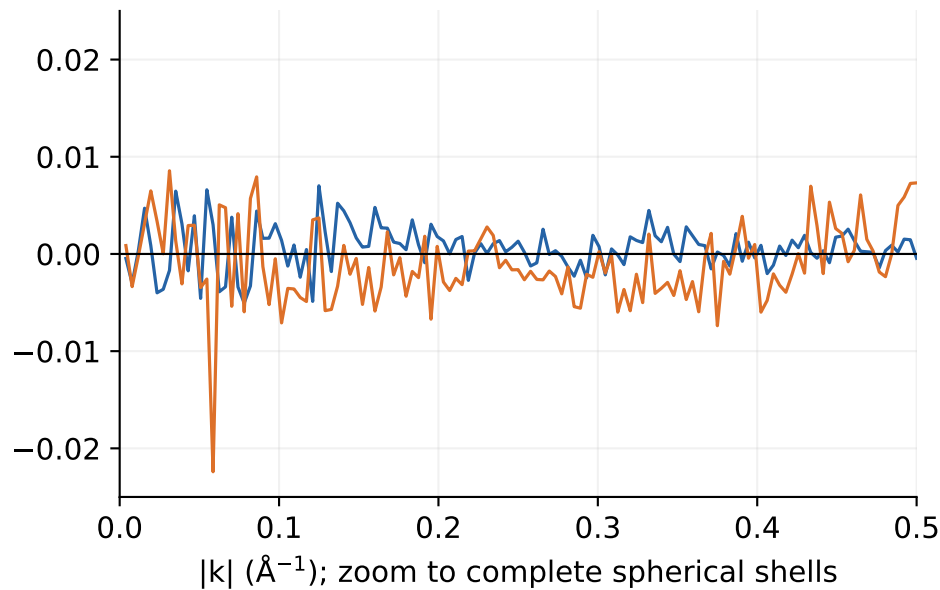
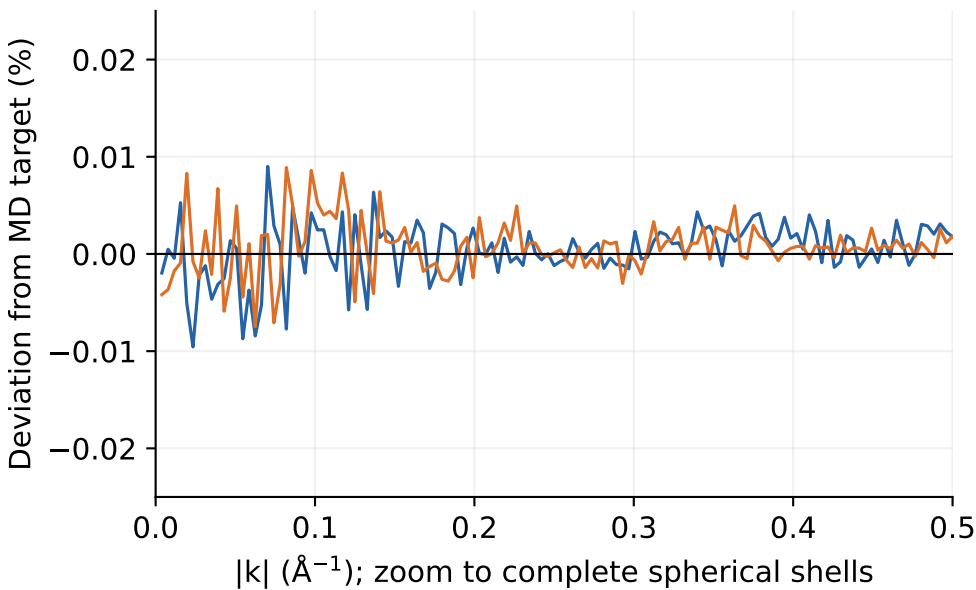
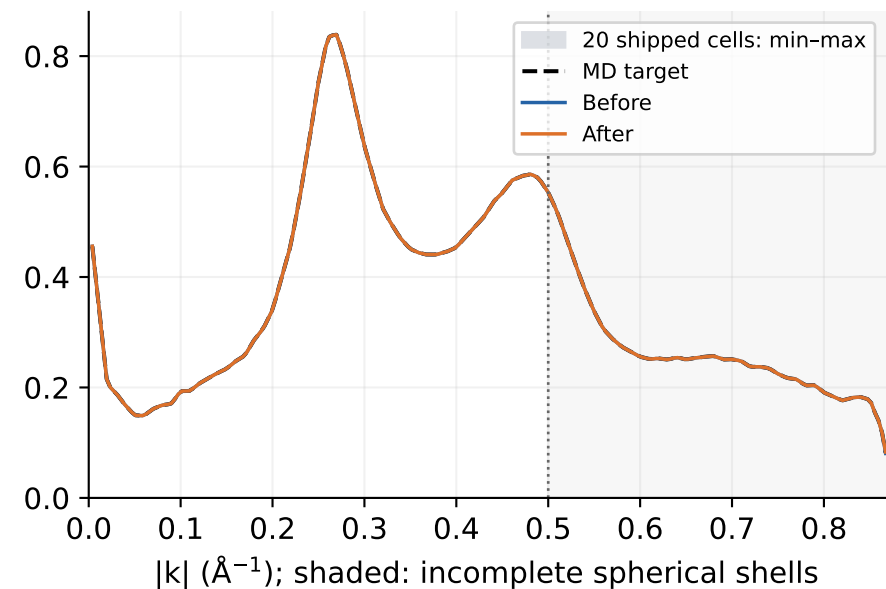
Seed 1000



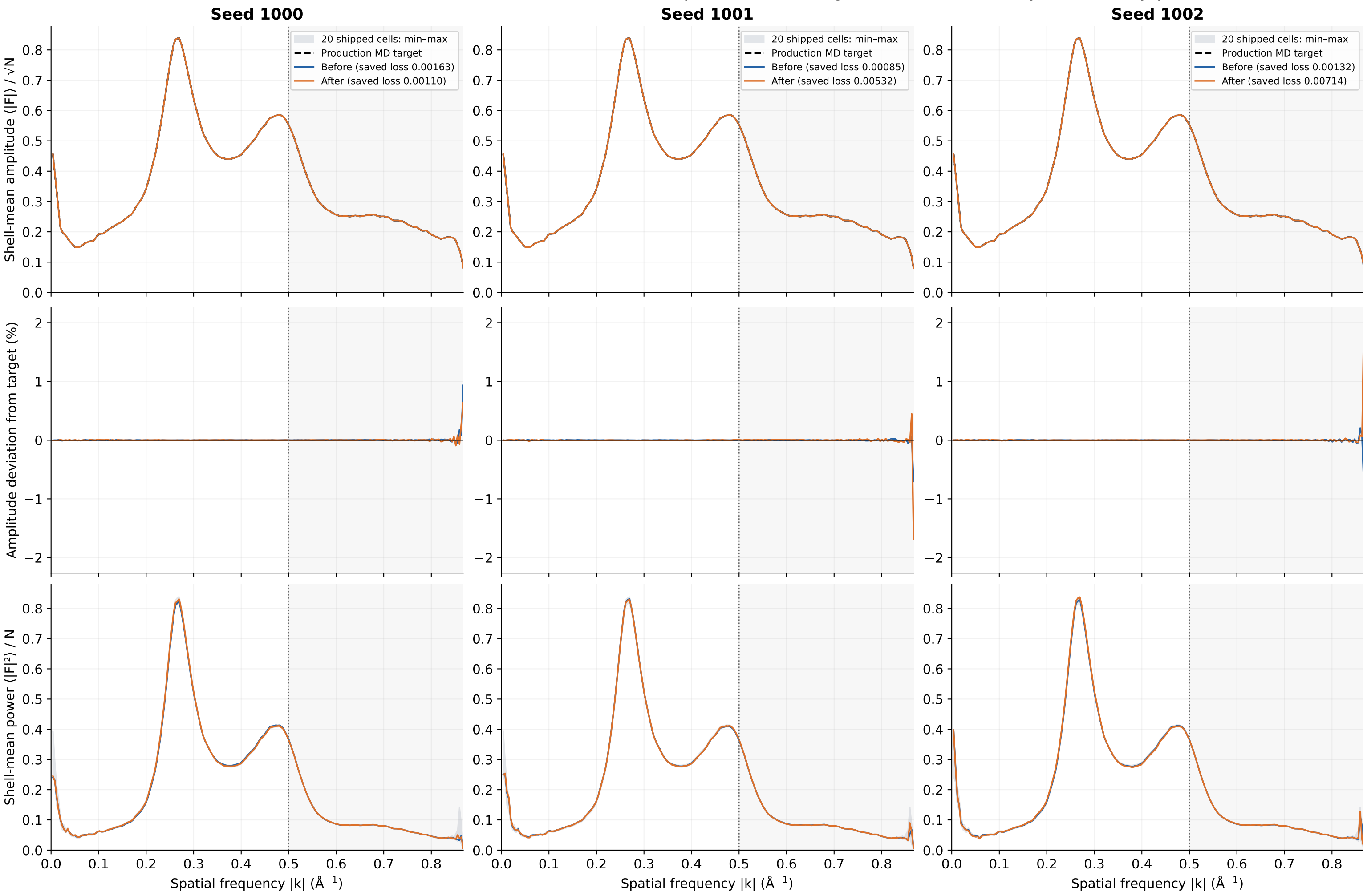
Seed 1001



Seed 1002

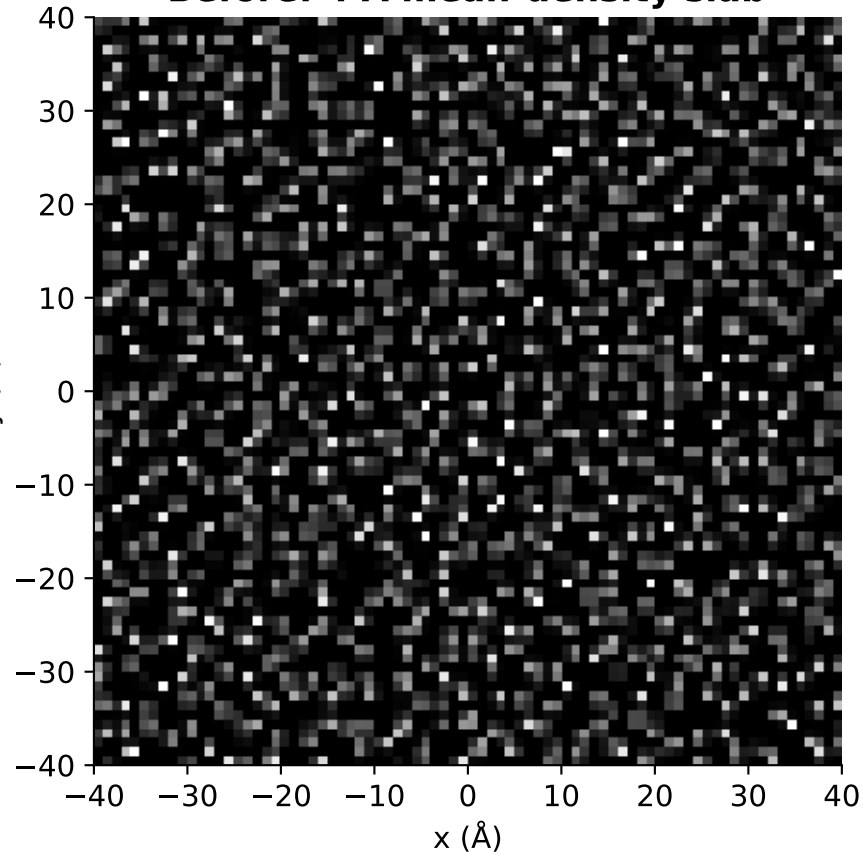


256³ at 1 Å/voxel · 527,178 molecules · DC omitted from plots · shaded region: shells extend beyond axial Nyquist

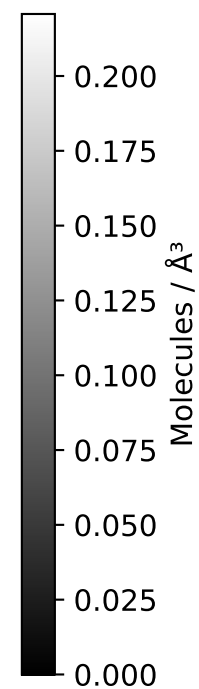
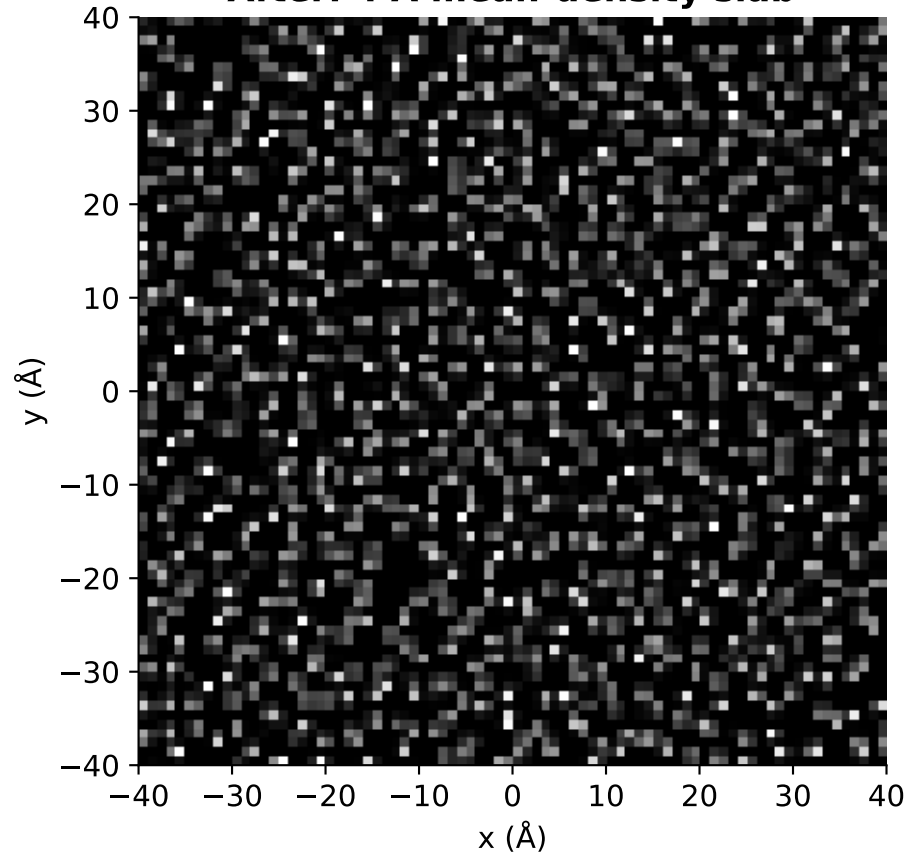


Seed 1000: saved ice cells before and after
Common scales across all seeds · same spatial slab / Fourier plane · different optimizer trajectories

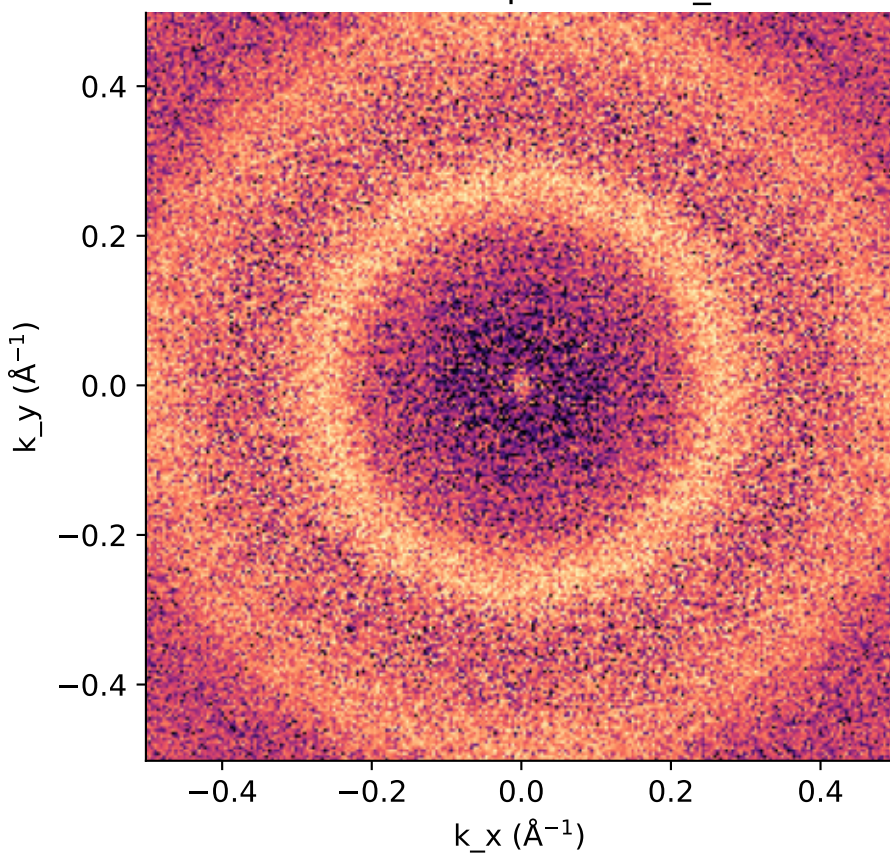
Before: 4 Å mean-density slab



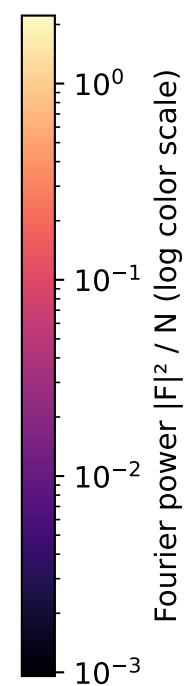
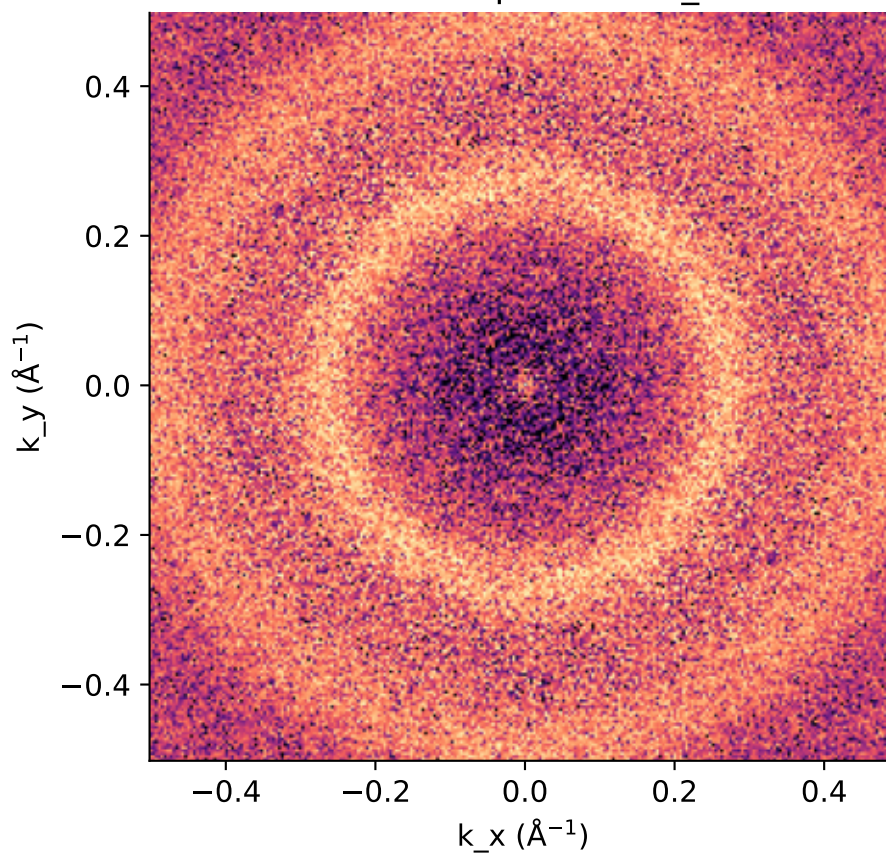
After: 4 Å mean-density slab



Before: Fourier power at $k_z = 0$

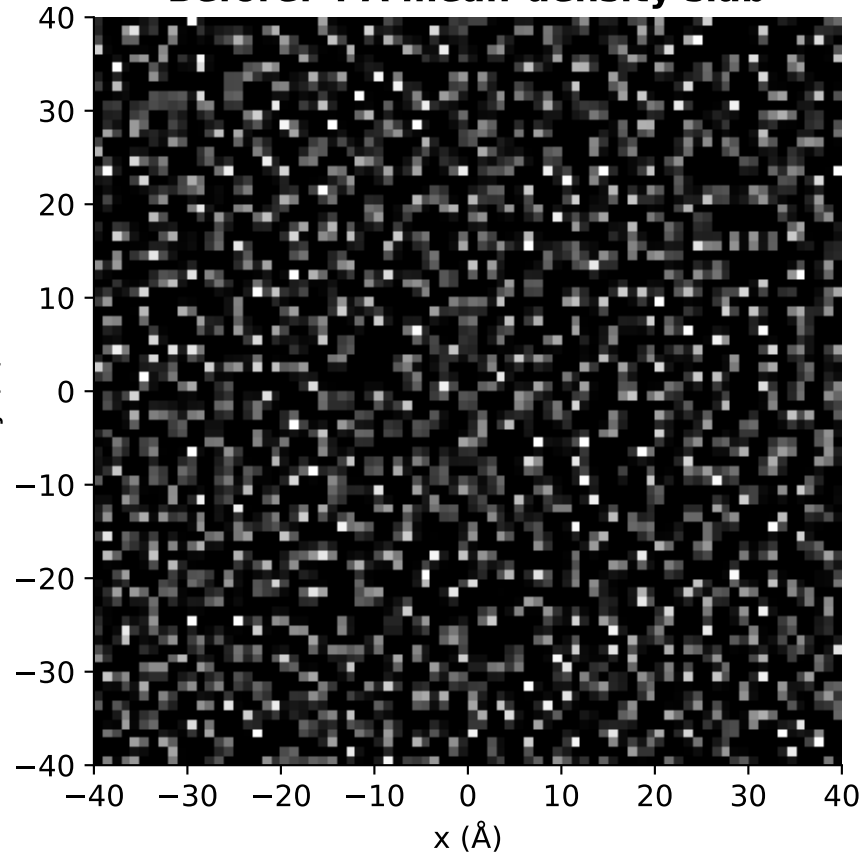


After: Fourier power at $k_z = 0$

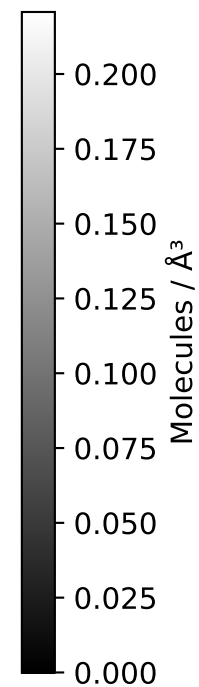
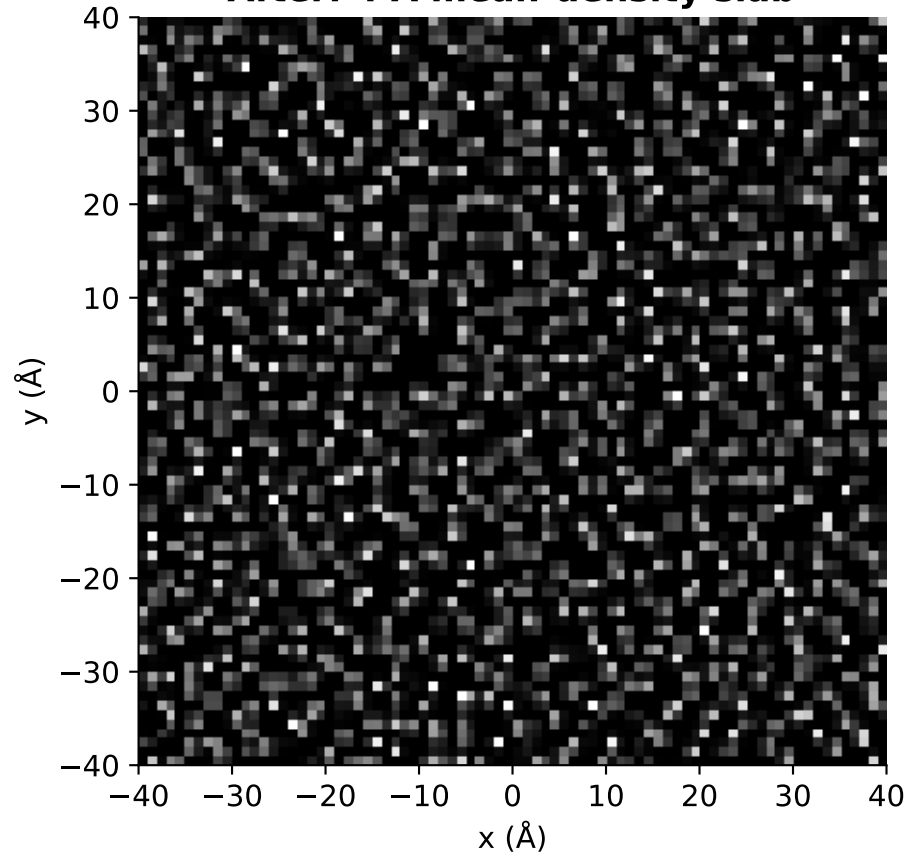


Seed 1001: saved ice cells before and after
Common scales across all seeds · same spatial slab / Fourier plane · different optimizer trajectories

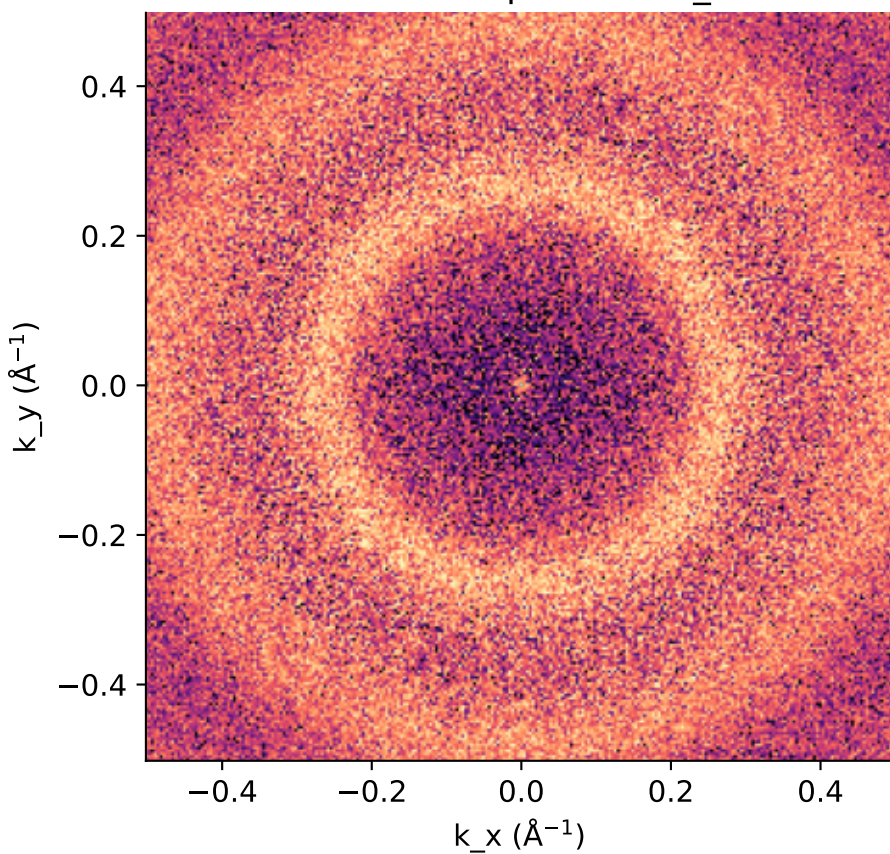
Before: 4 Å mean-density slab



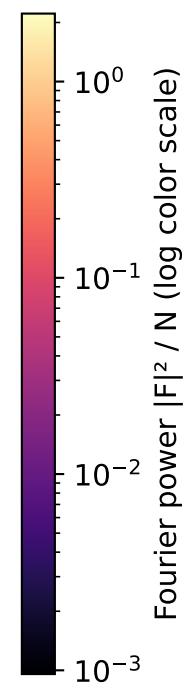
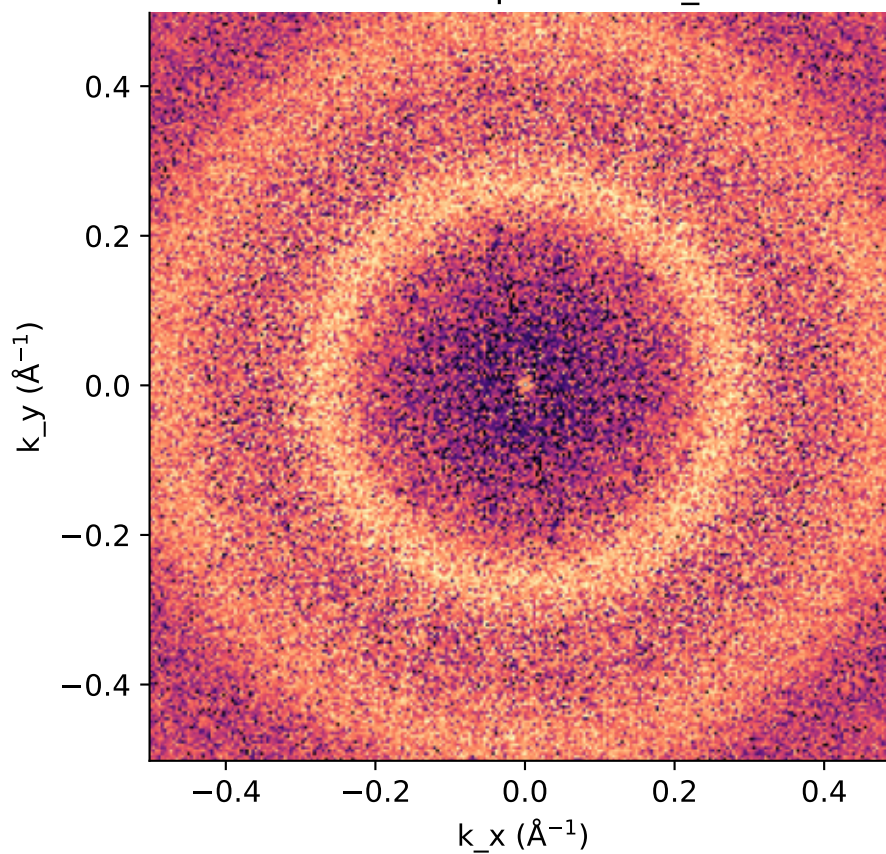
After: 4 Å mean-density slab



Before: Fourier power at $k_z = 0$

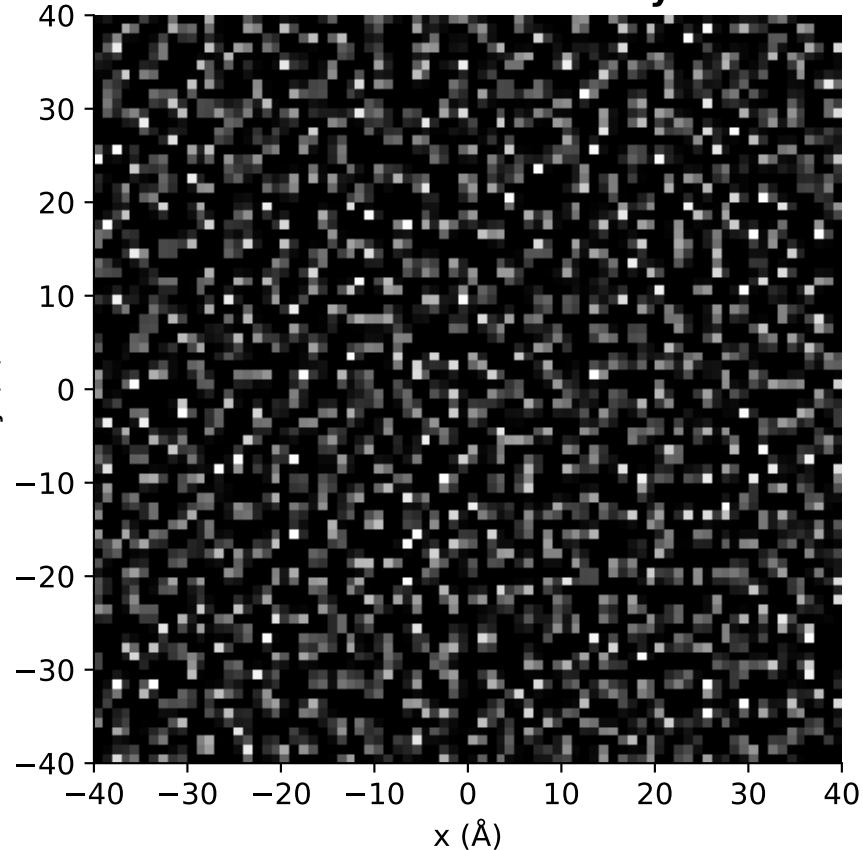


After: Fourier power at $k_z = 0$

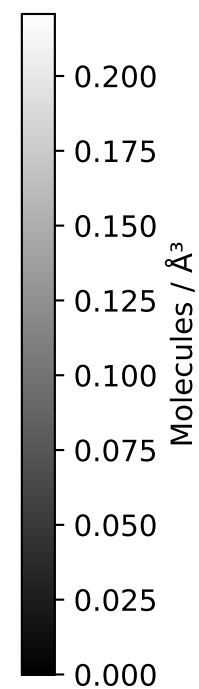
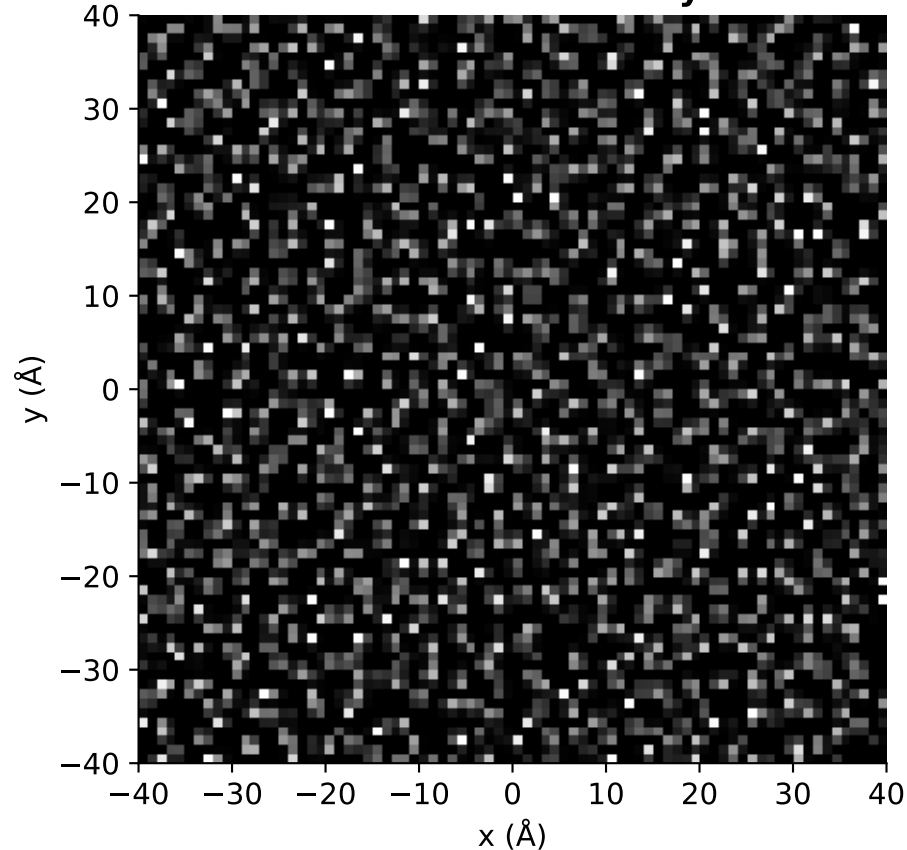


Seed 1002: saved ice cells before and after
Common scales across all seeds · same spatial slab / Fourier plane · different optimizer trajectories

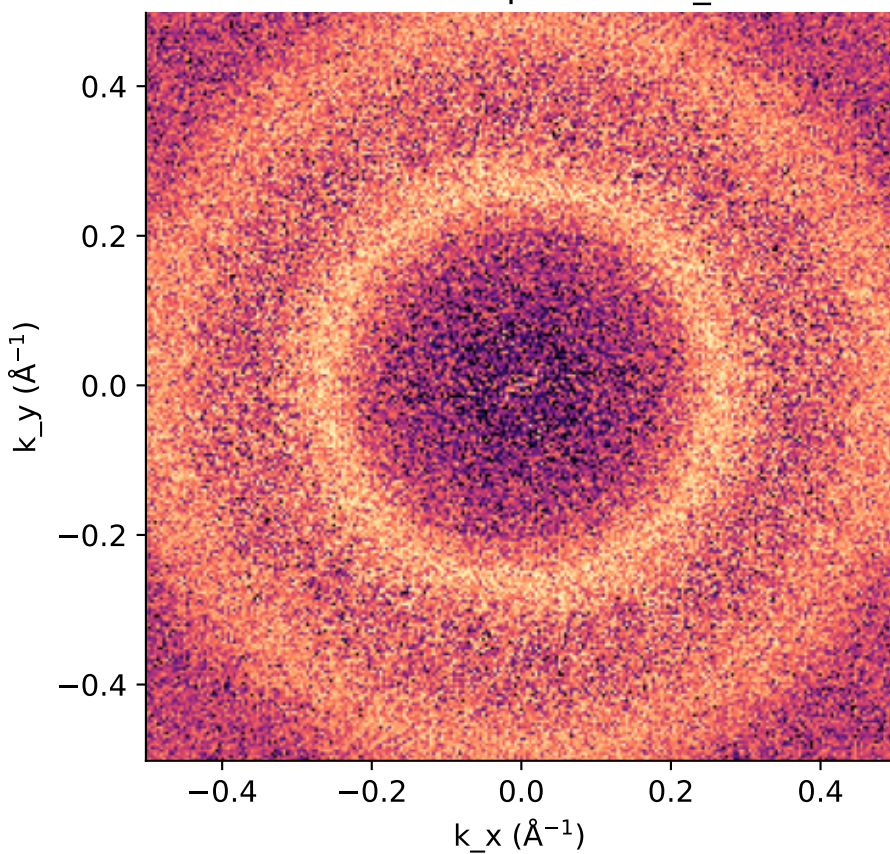
Before: 4 Å mean-density slab



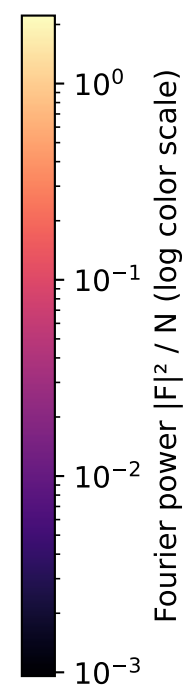
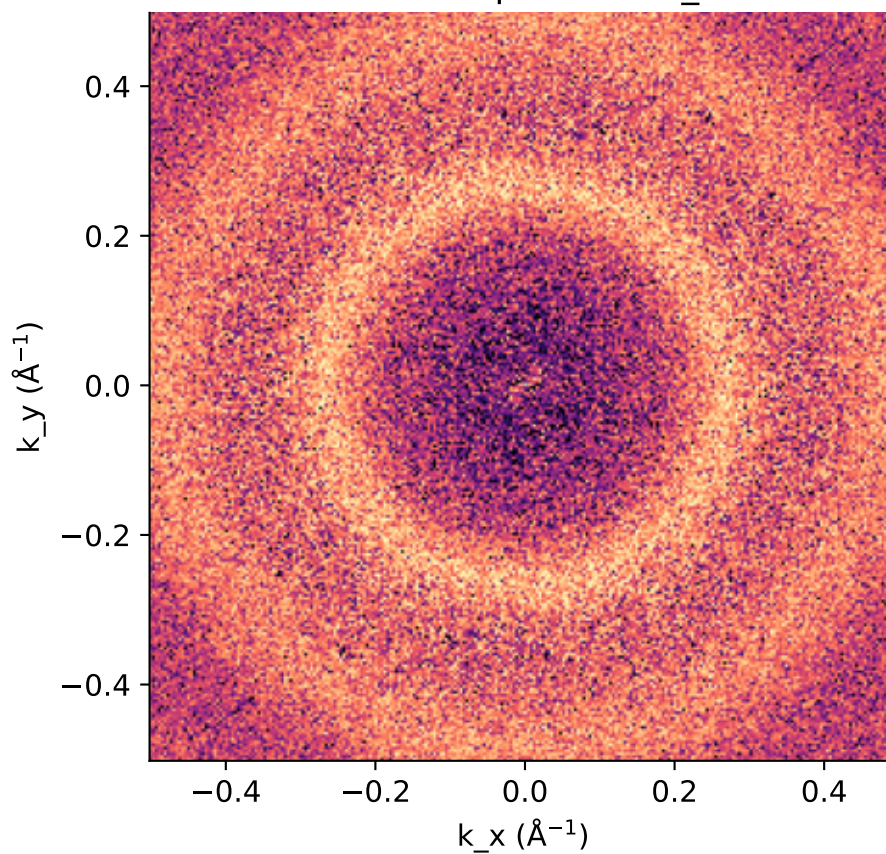
After: 4 Å mean-density slab



Before: Fourier power at $k_z = 0$

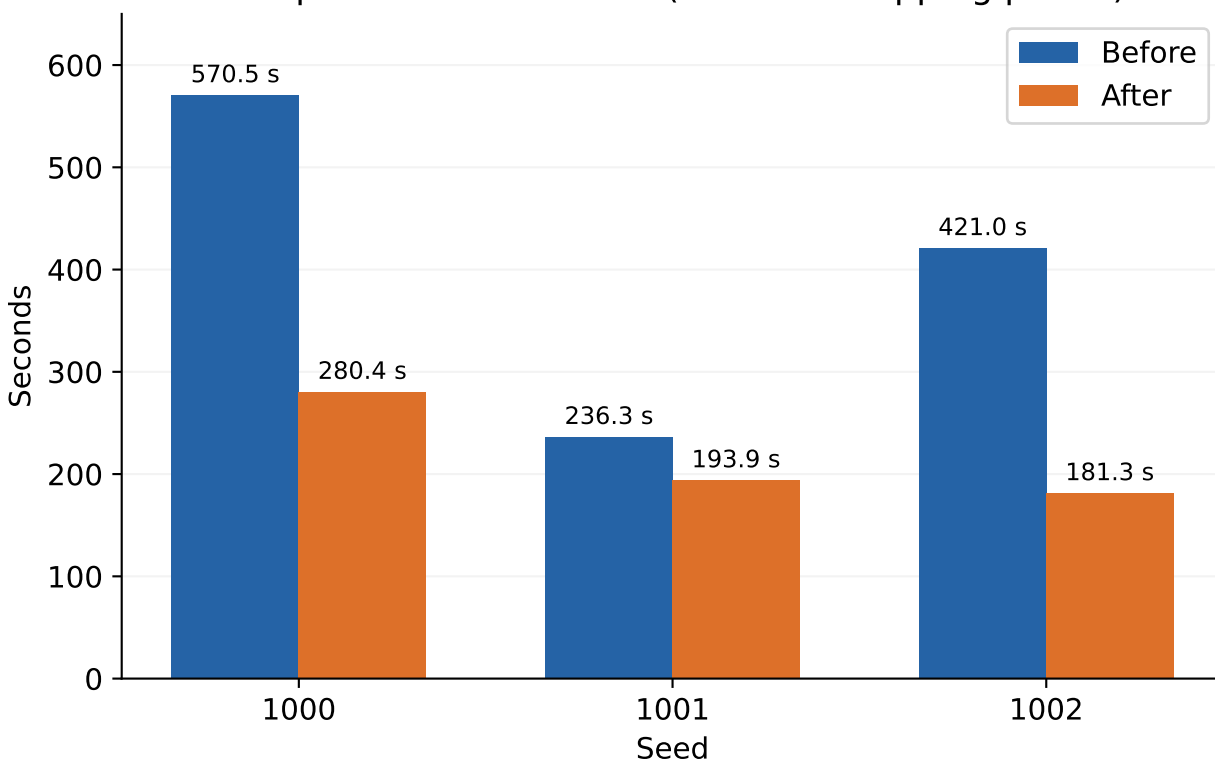


After: Fourier power at $k_z = 0$

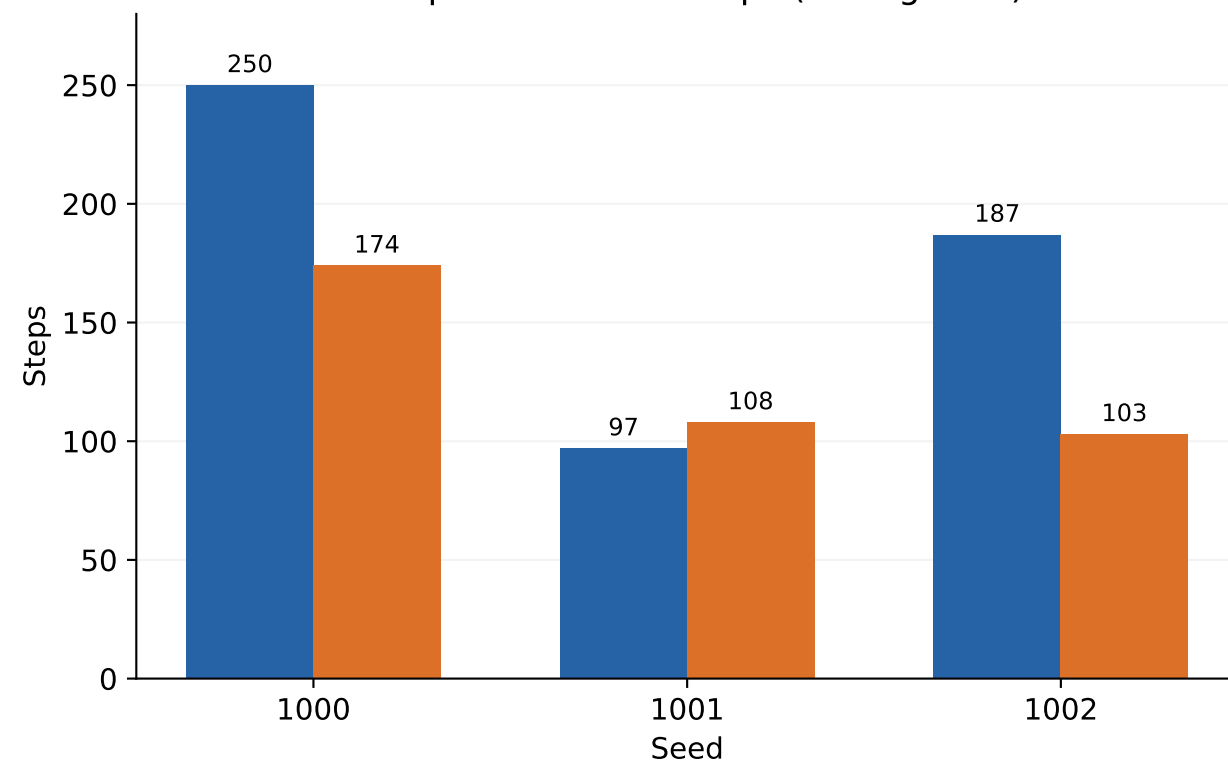


Measured NVIDIA L40 performance at production scale
 $256^3 \cdot dx = 1 \text{ \AA} \cdot 527,178$ molecules $\cdot$ unchanged dtypes and optimization recipe

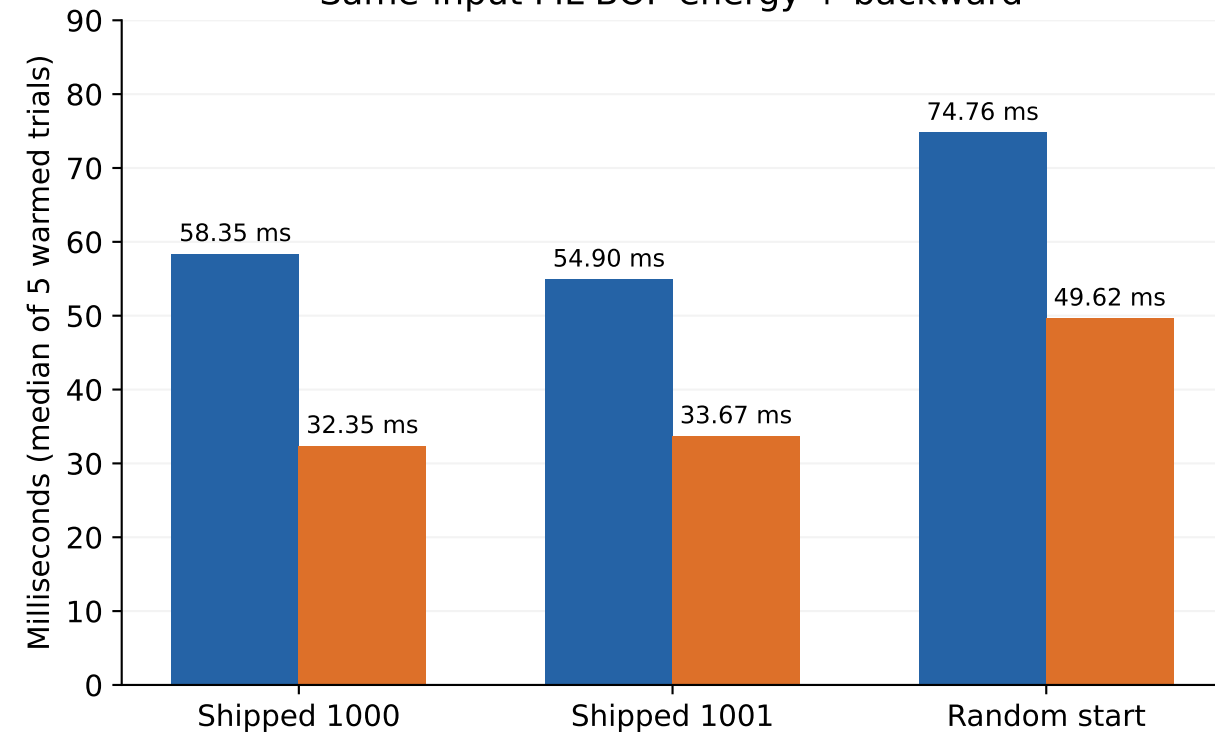
Complete build wall time (different stopping points)



Completed L-BFGS steps (ceiling: 250)



Same-input ML-BOP energy + backward



Saved-coordinate quality (lower loss is better)

